MODULE 5 — Evaluation: Metrics, Thresholds, and Decision Proxies

What You Measure Is What You Optimize

Terminology

- **Value judgement** - an explicit choice about which kinds of mistakes matter more than others.
  - In other words, a value judgement is simply a choice about which mistakes you are more willing to live with.

Why This Module Matters

Most data science education treats evaluation as a **performance comparison problem**:

- Which model is more accurate?
- Which metric is higher?

AI-native systems treat evaluation as a **decision control problem**:

- Which mistakes are rewarded?
- Which users are acted on?
- Which risks are tolerated?

Evaluation is not neutral. It determines **how the system behaves in the real world**.

Takeaways

- "Quietly reward" is referring to what happens AFTER the model has been validated & approved
  - At some point, the developed model is going to be implemented into a business process
  - It is at that point decisions will be made on the outcome of the model
- In the same way all the other pipeline steps introduce bias aligned with target decision-making to address uncertainty, choosing evaluation metrics does the same
  - Here it's deciding which metrics to use to summarize how well a model captures real world representation in its learned trends & outcome

Anchor Back to the Decision

We are still deciding:

Which customer cohorts should receive churn interventions under budget and operational constraints, with uncertainty about whether observed behavior reflects true churn risk or temporary noise.

Evaluation must reflect:

- **irreversible costs**
- **asymmetric risk**
- **operational capacity**
- **trust and explainability**

Core Reframe (Memorize This)

A metric is a proxy for a value judgment about which mistakes you are willing to make.

If you don't choose the value judgment intentionally, the metric will choose it for you.

Core Reframe (Memorize This)

Metrics express value judgments. Thresholds enforce policy.

A model produces scores. A threshold turns scores into actions.

Evaluation is incomplete until both are specified.

Why Accuracy Is Usually the Wrong Metric

Accuracy assumes:

- all errors are equally costly
- class balance matters more than consequences
- false positives and false negatives are symmetric

None of these assumptions hold in most real decisions.

A system can be highly accurate and still:

- waste resources
- erode trust
- miss critical cases

Error Types as Business Outcomes

Think in **decisions**, not labels:

Outcome	What the System Did	Business Meaning
True Positive	Correct intervention	Cost incurred, value protected
False Positive (Type I Error)	Unnecessary intervention	Cost + trust erosion
False Negative (Type II Error)	Missed intervention	Lost customer / opportunity
True Negative	No action	No cost, no impact

Metrics decide **which row dominates behavior**.

Strategy A vs Strategy B → Metric Implications

Your earlier work now constrains evaluation.

🔹 Strategy A — Stability / Precision / Cost Control

Primary concern: Avoid unnecessary, costly interventions.

Metrics that align:

- Precision
- False positive rate
- Cost-weighted loss
- Precision at top-K

Metrics to treat cautiously:

- Recall
- Overall accuracy

Why: It is acceptable to miss some churn if cost and trust are protected.

🔹 Strategy B — Sensitivity / Early Warning / Recall

Primary concern: Avoid missing true churn signals.

Metrics that align:

- Recall

- Sensitivity
- Coverage
- Time-to-detection

Metrics to treat cautiously:

- Precision
- FPR

Why: It is acceptable to over-intervene if early detection matters.

Thresholding: From Scores to Decisions

Models do not make decisions. They estimate risk.

Thresholds determine who is acted on.

Two systems using the same model can behave very differently depending on threshold choice.

Common Thresholding Strategies (AI-Native View)

🔹 Risk-Tolerance Thresholding

- Act only when confidence exceeds a minimum acceptable level
- Common when actions are costly or irreversible
- Naturally aligns with precision-focused systems

🔹 Capacity-Constrained Thresholding

- Threshold set so actions do not exceed operational limits
- Common in screening and triage systems
- Often paired with recall-focused objectives

🔹 Cost-Based Thresholding

- Threshold chosen to minimize expected cost
- Requires explicit cost asymmetry
- Aligns with cost-weighted loss, not balanced metrics

🔹 Tiered Thresholding (Advanced)

- Multiple thresholds:
  - high confidence → automatic action
  - medium confidence → human review
  - low confidence → no action

Thresholding is **policy**, not math.

Takeaways

- Don't think of threshold strategies go hand-in-hand with metrics for a given decision process.
  - They are related, but threshold strategy selection is not derived from evaluation metric selection
  - They are chosen in parallel, not sequentially

Common Thresholding Failure Modes

- Using 0.5 by default without justification
- Choosing thresholds solely to maximize a metric
- Ignoring operational capacity
- Treating thresholds as static over time
- Automating irreversible actions without human override

These failures rarely show up in offline evaluation.

Evaluation Is Not One Number

AI-native evaluation includes:

- multiple metrics
- performance by cohort and segment
- worst-case behavior
- threshold sensitivity analysis
- stability over time

A single headline metric hides risk.

The Correct Mental Model

Think of it like this:

The decision defines three independent design choices

All three must align, but none *determines* the others.

1. **Evaluation Metrics**
   → What behavior is rewarded during evaluation
   → Which errors look “good” or “bad” on paper
1. **Threshold Strategy**
   → How scores become actions in the real world
   → Who gets acted on, how often, and under what confidence
1. **Decision Constraints**

→ Cost, capacity, reversibility, trust, governance

The **decision** constrains all three.

Why This Distinction Matters

If you treat thresholding as *derived from metrics*, you get systems that:

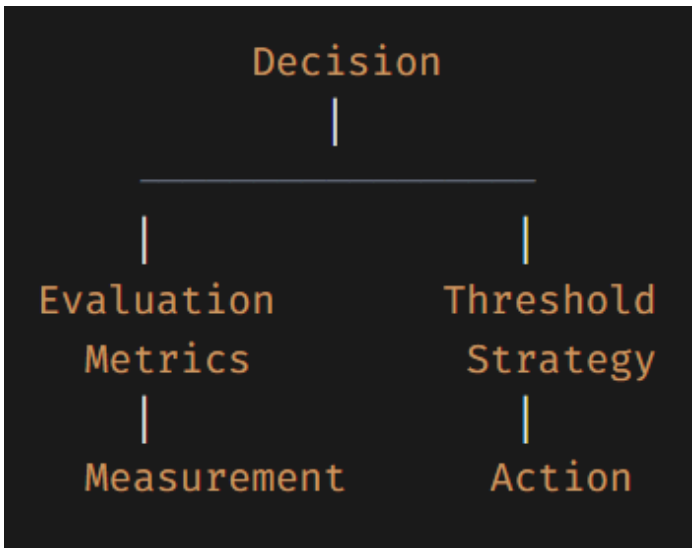
- optimize nicely offline
- behave poorly in production

If you treat metrics as *derived from thresholds*, you get systems that:

- look inconsistent
- are hard to reason about
- confuse stakeholders

AI-native systems avoid both mistakes by anchoring **both to the decision**.

A Useful Visual (Conceptual)



Neither box feeds the other. Both serve the decision.

The Clarifying Nuance (Important)

While **selection is independent**, **validation is coupled**.

That means:

- You choose metrics and thresholds independently
- But you must **check that they don't contradict each other**

Example of contradiction:

- Metric rewards recall
- Threshold enforces extreme conservatism

That system will:

- look good offline
- underperform operationally

So the relationship is:

Independent choice, coupled sanity-check

One-Sentence Anchor (Lock This In)

Metrics decide how performance is judged; thresholds decide how decisions are made — both must align to the same decision, but neither chooses the other.

You've internalized one of the most subtle (and most important) ideas in applied data science.

Prompt Engineering Pattern (Critical)

When working with AI on evaluation, use:

"Evaluate this system using metrics and thresholds aligned to decision risk, explicitly showing which errors are being favored and which are being sacrificed."

Never ask:

"What's the best metric or threshold?"

🧪 Module 5 Mini-Exercise — Metrics *and* Thresholds as Policy

Step 1 — Choose a Decision Strategy

Select one:

- Strategy A (costly, conservative intervention)
- Strategy B (early warning, recall-focused)

Step 2 — Select Evaluation Metrics

Choose 2–3 metrics aligned with the strategy.

For each metric:

1. Which errors does it reward or punish?
2. Which failure does it make easier to miss?

Step 3 — Choose a Thresholding Strategy

Select one:

- risk-tolerance
- capacity-constrained
- cost-based
- tiered

Explain:

- what constraint it enforces
- what risk it accepts

Step 4 — Explicitly Reject One Metric or Threshold

Complete the sentence:

"We explicitly reject this metric or threshold because…"

Tie the rejection to:

- decision risk
- irreversibility
- trust
- operational limits

Why This Module Is Career-Defining

Evaluation is where:

- values become numbers
- policy becomes code
- responsibility is assigned

Teams remember who:

- chose the metric
- justified the threshold
- warned about blind spots

End-of-Module Takeaway

A system is not defined by its model,but by the metrics and thresholds that decide who pays the price.

Relevant Articles

Confusion Matrix: How to Use It & Interpret Results [Examples]

- https://www.v7labs.com/blog/confusion-matrix-guide?utm_source=chatgpt.com

Deep Quantile Regression for Unsupervised Anomaly Detection in Time-Series

- https://www.researchgate.net/figure/A-trade-off-between-precision-and-recall-used-for-setting-up-a-threshold-value_fig5_354948827

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Introduction to Example-Dependent Cost-Sensitive Classification

- https://albahnsen.github.io/CostSensitiveClassification/Intro.html?utm_source=chatgpt.com

COVID-19 Patient Health Prediction Using Boosted Random Forest Algorithm

- https://www.researchgate.net/figure/Evaluation-metrics-for-decision-tree_fig3_342410608

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